

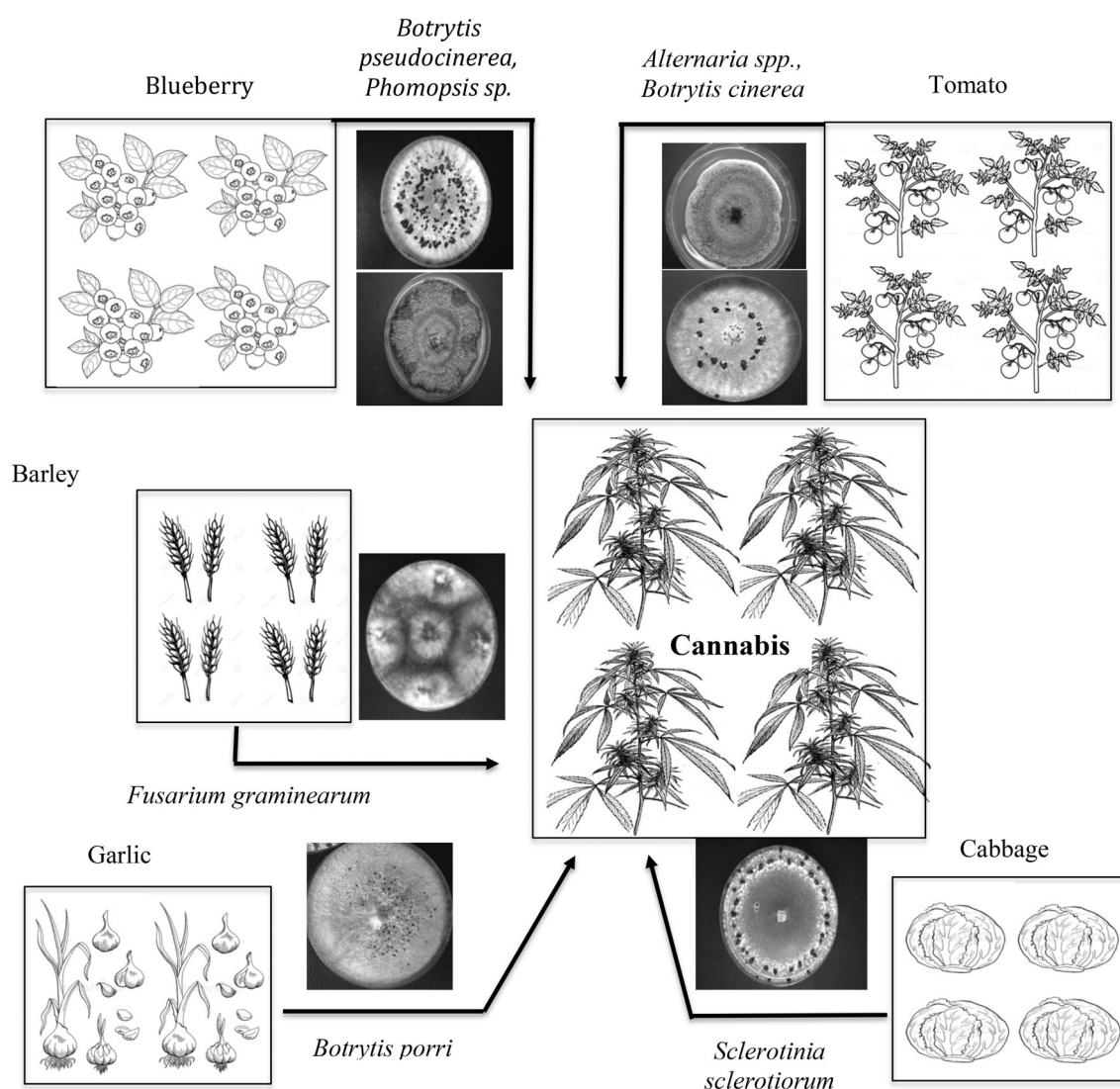


Figure 4. Pathogen inoculum sources for cannabis and hemp plants can originate from neighboring fields cropped to blueberry, tomato, barley, garlic and cabbage, which release inoculum to cause infection.

preventing infection or symptom development, reducing secondary inoculum production and spread, and reducing survival of the pathogen. These approaches are summarized in Table 1 and will be discussed with reference to specific pathogens that can be mitigated to prevent infection of cannabis or hemp plants.

8.1 Prevention of inoculum introduction

The prevention of inoculum introduction is critical for producers starting a new crop. This can be ensured through testing of stock plants for the presence of pathogens prior to vegetative cuttings being taken, disinfecting seeds and equipment to eradicate pathogen propagules, ensuring that planting materials, substrates and water sources are free from pathogen inoculum, and ensuring that cannabis and hemp crops are not planted in fields with a prior disease outbreak in the preceding crop, or adjacent to other crops that could be a source of inoculum. Methods for testing of plants to ensure they and the planting materials are free from pathogens are widely available and several commercial laboratories can provide detection services. PCR methods for detection of many species reported to infect cannabis and hemp, including

Fusarium and *Pythium*,^{21,69,70} as well as *Golovinomyces* species,^{41,71} have been described. *Hop latent viroid* can be detected using molecular techniques.^{22,23,72} Methods for detecting seed-borne pathogens on cannabis or hemp seeds can be adapted from the vast literature on seed-borne pathogens.^{65,73} Since there is currently no certification program for cannabis or hemp in Canada, licensed producers cannot be guaranteed disease-free planting materials. As a result, spread of *Fusarium* through cuttings provided by propagators, for example, is not uncommon, resulting in introduction of the pathogen into new areas. The powdery and downy mildew pathogens, and *Hop latent viroid*, can also be introduced on diseased cuttings, as these obligate pathogens require a living host for reproduction and survival. Many producers rely on propagators that may not be performing due diligence in ensuring that materials sold for distribution are certified pathogen-free. The development of tissue culture methods for cannabis and hemp has shown some success with regard to regeneration of shoots from nodal explants for micropropagation.^{74,75} While several commercial companies are now producing plantlets, this approach needs to be augmented